Exoplanet Hunters: Transit Detection and Characterisation of Planets Around STAC-1, STAC-2 and STAC-3

Operation Dim Star — STAC WARS, Problem Statement PS 3 (Astronomy)
Analysis Report

***Abstract*—Relative brightness time series of three stars (STAC-1, STAC-2 and STAC-3) have been analysed for transits. All three stars exhibit periodic, constant depth dips, with orbital periods of 4.25 d, 80 d and 37.5 d and transit depths of 0.0030, 0.0010 and 0.00088 respectively. Using $\delta \approx (R_p/R_s)^2$ the planetary radii are found to be 5.98, 2.76 and 1.62 R_E (Neptune sized, Super-Earth, Super-Earth). Kepler's Third Law gives the planets orbital distances of 0.051, 0.337 and 0.174 AU. STAC-3b is the most promising candidate for habitability follow-up, although it is within the simplified Habitable Zone distance and is not claimed to be habitable.**

***Index Terms*—exoplanets, transit photometry, transit depth, Kepler's Third Law, Habitable Zone.**

I. INTRODUCTION

A planet transiting in front of its host star will block a small fraction of the star's light, resulting in a brief, periodic dip in the observed brightness known as a transit [1],[3]. This report analyses telescope data of three stars to (i) detect transiting planets, (ii) determine their radii and orbital distances and (iii) select the most promising candidate for further habitability work.

II. DATA AND ASSUMPTIONS

Stellar properties are given in Table I. Brightness is relative to the out of transit level so $F_{normal} = 1.0000$. Constants used: R_{Sun} = 696,000 km, R_E = 6,371 km, 1 AU = 149.6×10^6 km, 1 yr = 365.25 d.

TABLE I. STELLAR PROPERTIES

Property	STAC-1	STAC-2	STAC-3
Mass (M_{Sun})	1.00	0.80	0.50
Radius (R_{Sun})	1.00	0.80	0.50
Luminosity (L_{Sun})	1.00	0.40	0.08
T_{eff} (K)	5800	5000	3800

Assumptions: (1) each periodic dip is due to a single planet on a circular orbit; (2) the planet mass is negligible compared to the star mass; (3) the period is the mean spacing between the observed dips as the sampling interval (1, 20 and 10 days) may miss the true transit time; (4) the transit depth is the mean of the observed dips; (5) planet classes are Earth sized (<1.25 R_E), Super-Earth (1.25 to 3 R_E), Neptune sized (3 to 8 R_E) and Jupiter sized (>8 R_E); (6) limb darkening, impact parameter and planetary albedo are ignored.

Note on the problem statement: the problem statement has $\delta \approx (R_s/R_p)^2$. Physically the fraction of the stellar disc blocked is $(R_p/R_s)^2$ so the latter form is used here (the former would give planets larger than their stars).

III. PART A: DETECTION OF EXOPLANETS

The transit depth is defined as

$$\delta = (F_{normal} - F_{transit})/F_{normal} \qquad (1)$$

Figure 1 shows the three light curves with the dip points highlighted.

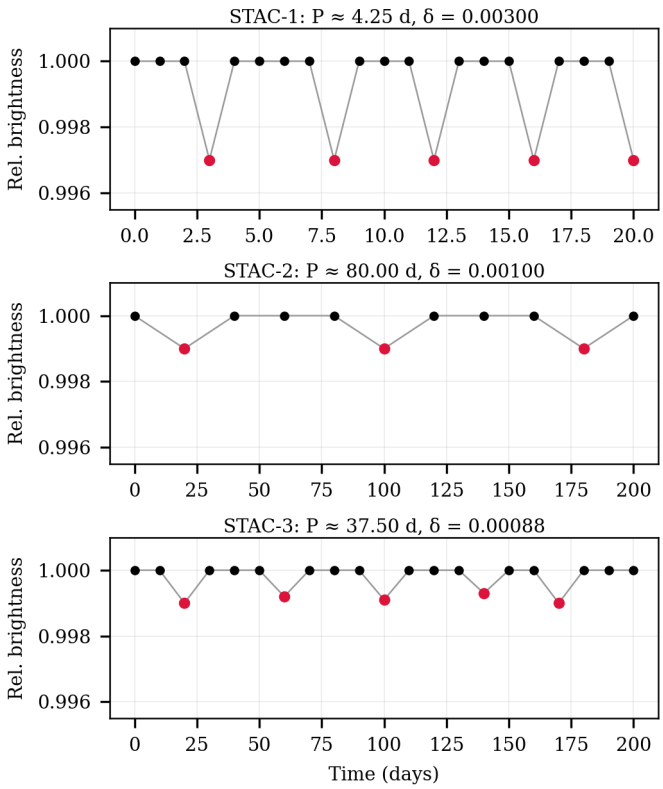


Fig 1. Relative brightness of STAC-1, STAC-2 and STAC-3. Red points mark the transit dips.

A. STAC-1

Dips (0.9970) occur at t = 3, 8, 12, 16, 20 d. The gaps are 5, 4, 4, 4 d. With one sample per day the true transit time lies somewhere within a day of each sample so the best period estimate is the mean spacing:

$$P_1 = (20 - 3)/4 = 4.25 \text{ d} \qquad (2)$$

$$\delta_1 = (1.0000 - 0.9970)/1.0000 = 0.0030 \qquad (3)$$

B. STAC-2

Dips (0.9990) occur at t = 20, 100, 180 d, equally spaced by 80 d. Samples are 20 d apart, so the transit may fall between samples, but the repeated equal spacing gives

$$P_2 = (180 - 20)/2 = 80 \text{ d}, \delta_2 = 1 - 0.9990 = 0.0010 \qquad (4)$$

C. STAC-3

Dips at t = 20, 60, 100, 140, 170 d with depths 0.0010, 0.0008, 0.0009, 0.0007, 0.0010. The first four are 40 d apart, the last is only 30 d after the fourth. With sampling every 10 d the true transit times are uncertain by as much as ~10 d so the mean spacing is used:

$$P_3 = (170 - 20)/4 = 37.5 \text{ d} \qquad (5)$$

$$\delta_3 = (0.0010+0.0008+0.0009+0.0007+0.0010)/5 = 0.00088 \qquad (6)$$

A strict 40 d period only fits the first four dips. Using it changes a from 0.174 to 0.182 AU (about 4%, Section V) so the conclusions are unchanged.

TABLE II. DETECTED TRANSITS

Star	Dip times (d)	Period (d)	Depth δ
STAC-1	3, 8, 12, 16, 20	4.25	0.0030
STAC-2	20, 100, 180	80	0.0010
STAC-3	20, 60, 100, 140, 170	37.5	0.00088

Conclusion: all three stars host a transiting planet. **Why a periodic dip implies a planet:** an orbiting planet blocks the same fraction of the stellar disc each time it transits in front of the star so the dips have constant depth and occur at the orbital period. Random noise or starspots would not produce regularly spaced dips of equal depth.

IV. PART B: PLANET SIZE

From $\delta \approx (R_p/R_s)^2$:

$$R_p = R_s \sqrt{\delta} \qquad (7)$$

STAC-1 b: R_s = 1.00 × 696,000 = 696,000 km; √0.0030 = 0.0548; R_p = 696,000 × 0.0548 ≈ 38,121 km = 38,121/6,371 = 5.98 R_E.

STAC-2 b: R_s = 0.80 × 696,000 = 556,800 km; √0.0010 = 0.0316; R_p = 556,800 × 0.0316 ≈ 17,608 km = 2.76 R_E.

STAC-3 b: R_s = 0.50 × 696,000 = 348,000 km; √0.00088 = 0.0297; R_p = 348,000 × 0.0297 ≈ 10,323 km = 1.62 R_E.

TABLE III. PLANETARY RADII AND CLASSIFICATION

Planet	R_p (km)	R_p (R_E)	Class
STAC-1 b	38,121	5.98	Neptune-sized
STAC-2 b	17,608	2.76	Super-Earth
STAC-3 b	10,323	1.62	Super-Earth

STAC-2 b (2.76 R_E) is near the upper end of the Super-Earth range. Planets above ~1.6 R_E are typically volatile rich rather than purely rocky [4], so STAC-3 b at 1.62 R_E is the best candidate for a rocky composition.

V. PART C: ORBITAL DISTANCES

Kepler's 3rd law in solar units $P^2 = a^3/M$ is rearranged to give the orbital distance:

$$a = (P^2 M)^{1/3} \qquad (8)$$

STAC-1 b: P = 4.25/365.25 = 0.01164 yr; a = $(0.01164^2 \times 1.00)^{1/3}$ = 0.0513 AU = 0.0513 × 149.6×10^6 ≈ 7.68×10^6 km.

STAC-2 b: P = 80/365.25 = 0.2190 yr; a = $(0.2190^2 \times 0.80)^{1/3}$ = 0.337 AU ≈ 50.5×10^6 km.

STAC-3 b: P = 37.5/365.25 = 0.1027 yr; a = $(0.1027^2 \times 0.50)^{1/3}$ = 0.174 AU ≈ 26.0×10^6 km.

TABLE IV. ORBITAL DISTANCES

Planet	a (AU)	a (km)	a / 1 AU	Compared with Earth
STAC-1 b	0.051	7.68×10^6	0.051	19.5× closer
STAC-2 b	0.337	5.05×10^7	0.337	3.0× closer
STAC-3 b	0.174	2.60×10^7	0.174	5.7× closer

All three planets orbit much closer to their stars than Earth does to the Sun (1 AU). If STAC-3 b had P = 40 d then a = 0.182 AU (a change of ~4%).

VI. PART D: MOST PROMISING WORLD

The simplified habitable zone distance [2] is

$$d_{HZ} \approx \sqrt{(L/L_{Sun})} \text{ AU} \qquad (9)$$

As additional measures the stellar flux received (in Earth units) is $S = (L/L_{Sun})/a^2$ and the equilibrium temperature with zero albedo is $T_{eq} = T_{eff}\sqrt{(R_s/2a)}$ (planet assumed to re-radiate over its whole surface).

TABLE V. HABITABILITY COMPARISON

System	d_{HZ} (AU)	a (AU)	a / d_{HZ}	S (Earth=1)	T_{eq} (K)
STAC-1	1.000	0.051	0.05	379.3	1234
STAC-2	0.632	0.337	0.53	3.5	371
STAC-3	0.283	0.174	0.62	2.6	311

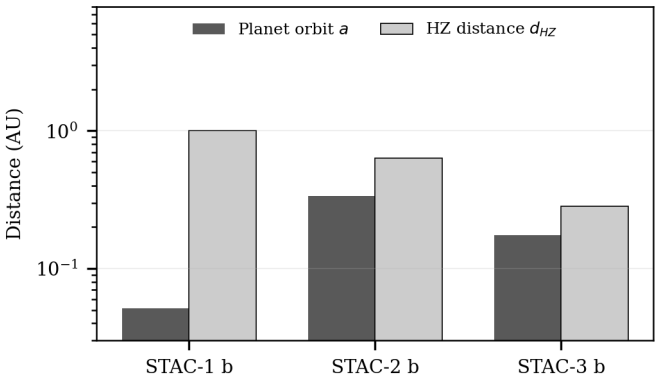


Fig. 2. Planet orbital distance compared to the simplified habitable zone distance of its star.

Choice: STAC-3 b. Evidence:

1) Closest to its habitable zone. a/d_{HZ} = 0.174/0.283 = 0.62 (the highest of the three: STAC-2 b: 0.53; STAC-1 b: 0.05). STAC-1 b is about 20 times too close.

2) Most Earth-like size. At 1.62 R_E it is the smallest of the three and most likely to be rocky. STAC-1 b (6.0 R_E) is probably a gas or ice giant and STAC-2 b (2.8 R_E) is likely volatile rich.

3) Lowest stellar flux. S = 2.6 (Earth = 1) vs 3.5 for STAC-2 b and ~379 for STAC-1 b.

4) Temperature and detection quality. The zero albedo T_{eq} is ~311 K for STAC-3 b (STAC-2 b: 371 K; STAC-1 b: 1234 K) and STAC-3 b has five separate transits recorded.

Caveat: STAC-3 b has $a/d_{HZ} < 1$ so it is inside the simplified habitable zone distance and is likely too warm. A planet near the habitable zone is not necessarily habitable (atmosphere, composition and magnetic environment are all unknown).

Suggested follow-up: transit transmission spectroscopy (e.g. with JWST). Starlight passing through the planet's atmosphere will have absorption features due to H_2O, CO_2, CH_4 or O_2 which will tell you whether an atmosphere is present and what it contains. A radial velocity mass measurement would also give you the density and hence whether the planet is rocky.

VII. CONCLUSION

All three stars have transiting planets. STAC-1b is a hot Neptune (6.0 R_E, P=4.25 d, a=0.051 AU). STAC-2b is a Super-Earth (2.8 R_E, P=80 d, a=0.337 AU). STAC-3b is a Super-Earth (1.6 R_E, P=37.5 d, a=0.174 AU) and is the most promising candidate for follow-up work (with the caveat that it is inside the habitable zone distance). The main uncertainties are in the period estimates (limited by the sampling interval) and the neglected impact parameter and limb darkening.

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